Exchange space expansion

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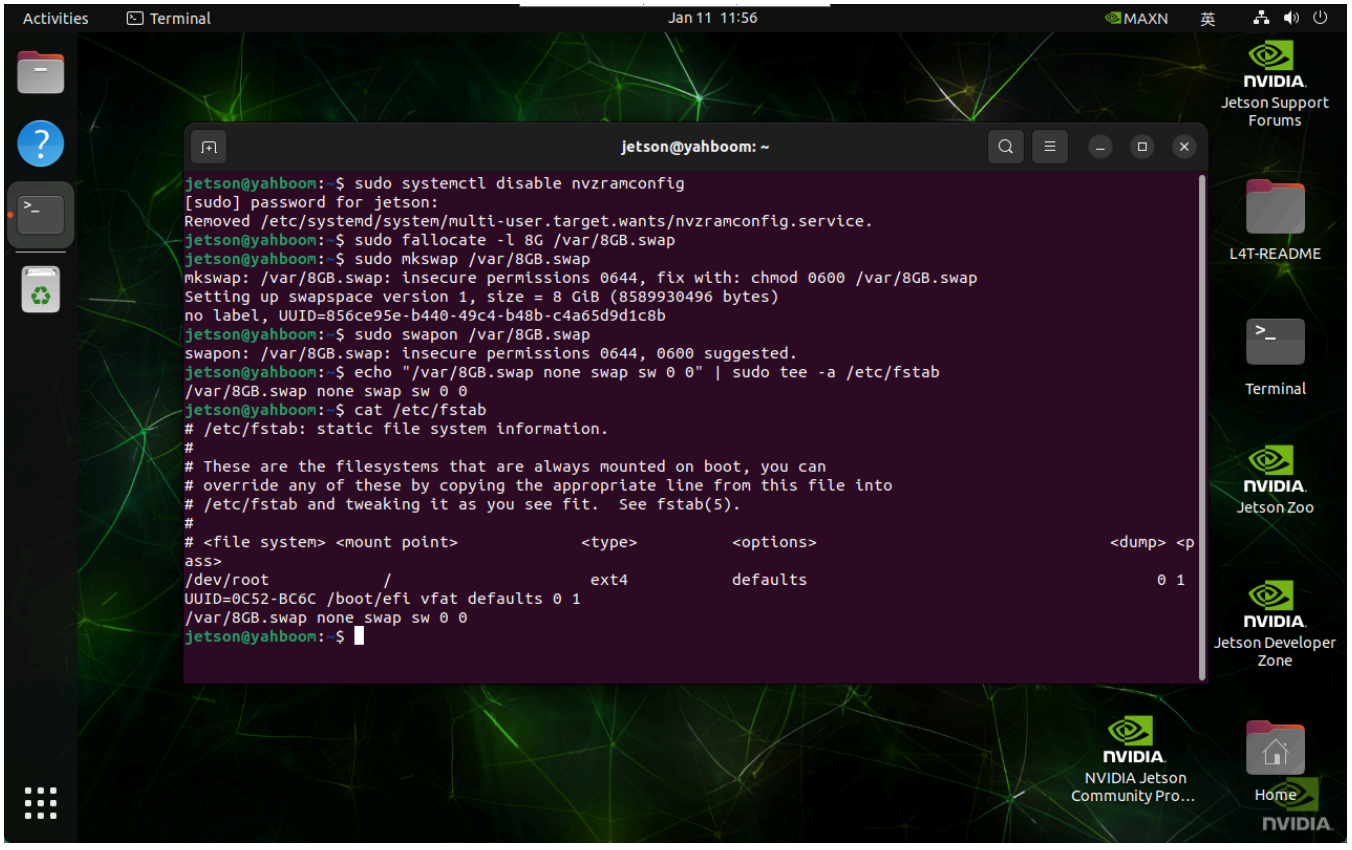
1. Exchange space

Exchange space is a mechanism that the operating system uses to extend the available memory. It keeps the system running when the memory is insufficient, avoiding program crashes or a system freeze!

Note: exchange space is accessed far more slowly than physical memory

2. Exchange space expansion

```
sudo systemctl disable nvzramconfig
sudo fallocate -l 8G /var/8GB.swap
sudo mkswap /var/8GB.swap
sudo swapon /var/8GB.swap
echo "/var/8GB.swap none swap sw 0 0" | sudo tee -a /etc/fstab
```



2.1. Disable the ZRAM swap configuration

Disable the ZRAM swap configuration on the Jetson device: ZRAM compresses the memory pages and stores them in memory, to reduce the reliance on the disk.

```
sudo systemctl disable nvzramconfig
```

2.2. Create the 8GB file

Use `fallocate` to create an 8GB file at the path `/var/8GB.swap`.

```
sudo fallocate -l 8G /var/8GB.swap
```

2.3. Set up the exchange space format

```
sudo mkswap /var/8GB.swap
```

2.4. Enable the exchange space

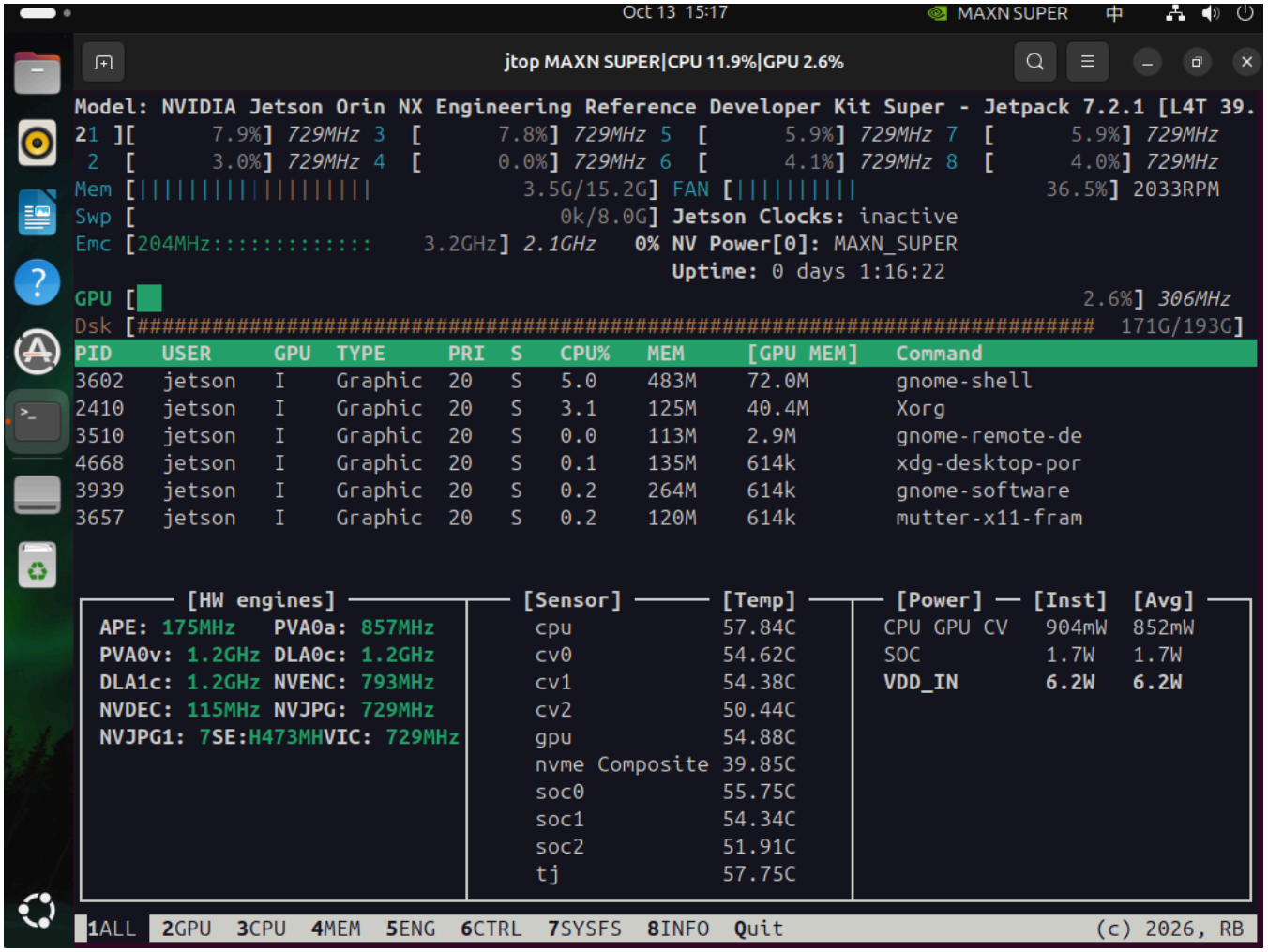
```
sudo swapon /var/8GB.swap
```

2.5. Start the exchange space automatically at boot

```
echo "/var/8GB.swap none swap sw 0 0" | sudo tee -a /etc/fstab
```

3. Verify the expansion

After the system reboots, the system exchange space has increased to 8GB:


Oct 13 15:17
MAXN SUPER
jtop MAXN SUPER|CPU 11.9%|GPU 2.6%
Model: NVIDIA Jetson Orin NX Engineering Reference Developer Kit Super - Jetpack 7.2.1 [L4T 39.
21][7.9%] 729MHz 3 [7.8%] 729MHz 5 [5.9%] 729MHz 7 [5.9%] 729MHz
2 [3.0%] 729MHz 4 [0.0%] 729MHz 6 [4.1%] 729MHz 8 [4.0%] 729MHz
Mem [3.5G/15.2G] FAN [36.5%] 2033RPM
Swp [0k/8.0G] Jetson Clocks: inactive
Emc [204MHz: 3.2GHz] 2.1GHz 0% NV Power[0]: MAXN_SUPER
Uptime: 0 days 1:16:22
GPU [2.6%] 306MHz
Dsk [171G/193G]
PID USER GPU TYPE PRI S CPU% MEM [GPU MEM] Command
3602 jetson I Graphic 20 S 5.0 483M 72.0M gnome-shell
2410 jetson I Graphic 20 S 3.1 125M 40.4M Xorg
3510 jetson I Graphic 20 S 0.0 113M 2.9M gnome-remote-de
4668 jetson I Graphic 20 S 0.1 135M 614k xdg-desktop-por
3939 jetson I Graphic 20 S 0.2 264M 614k gnome-software
3657 jetson I Graphic 20 S 0.2 120M 614k mutter-x11-fram
[HW engines]
APE: 175MHz PVA0a: 857MHz
PVA0v: 1.2GHz DLA0c: 1.2GHz
DLA1c: 1.2GHz NVENC: 793MHz
NVDEC: 115MHz NVJPG: 729MHz
NVJPG1: 7SE:H473MHVIC: 729MHz
[Sensor] [Temp]
cpu 57.84C
cv0 54.62C
cv1 54.38C
cv2 50.44C
gpu 54.88C
nvme Composite 39.85C
soc0 55.75C
soc1 54.34C
soc2 51.91C
tj 57.75C
[Power] [Inst] [Avg]
CPU GPU CV 904mW 852mW
SOC 1.7W 1.7W
VDD_IN 6.2W 6.2W
1ALL 2GPU 3CPU 4MEM 5ENG 6CTRL 7SYSFS 8INFO Quit (c) 2026, RB